# David Topping

Professor of the Digital Environment, University of Manchester · Director of Digital Worlds, Digital Futures

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I build computational models and digital tools for environmental science, with roots in atmospheric and aerosol science and a focus today on machine learning, AI systems, sensing and data infrastructure. I have published in journals including Nature, Nature Geoscience and Nature Communications, been PI or Co-I on grants totalling more than £30M, edit for two international journals, and co-founded a university spin-out. I advise UK government and research councils, including NERC, DEFRA and UKHSA, on the use of AI and data science in environmental science, and work with industry partners including NVIDIA.

## ACADEMIC APPOINTMENTS

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| 2025-present | **Director of Digital Worlds.** Digital Futures, University of Manchester |
| 2024-present | **Adjunct Professor.** Zhejiang University, China |
| 2021-present | **Professor of the Digital Environment.** University of Manchester |
| 2018-2024 | **Fellow.** The Alan Turing Institute |
| 2018-2021 | **Reader.** University of Manchester |
| 2017-2018 | **Senior Lecturer.** Centre for Atmospheric Science, School of Earth and Environmental Sciences, University of Manchester |
| 2013-2017 | **Senior Research Fellow.** Centre for Atmospheric Science, University of Manchester |

## EDUCATION

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| 2005 | **PhD, Modelling the hygroscopic properties of atmospheric aerosol particles.** UMIST. Supervisor Professor Hugh Coe. |
| 2001 | **BSc (Hons) Pure Physics, First Class.** UMIST |

## ADVISORY AND EXTERNAL ROLES

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| 2025-present | **Honorary Academic.** UK Health Security Agency |
| 2024-present | **International Advisor.** Hong Kong Baptist University, Digital Solutions programme |
| 2023-2028 | **Member, Science & Innovation Committee.** Natural Environment Research Council (NERC) |
| 2022-present | **Member, Air Quality Expert Group.** DEFRA |
| 2022-present | **Member, Programme Executive Group for Digital Solutions.** NERC |
| 2022-present | **Co-founding Director and Chief Technology Officer.** Urban360, a University of Manchester spin-out |

## JOURNAL EDITORSHIPS

| | |
|---|---|
| 2025-present | **Deputy Editor-in-Chief.** Big Data and Earth System (KeAi / Elsevier) |
| 2018-2025 | **Topical Editor.** Geoscientific Model Development (Copernicus) |
| 2012-2021 | **Co-editor.** Atmospheric Chemistry and Physics (Copernicus) |

## SELECTED RESEARCH FUNDING

*PI or Co-I on competitively awarded grants exceeding £30M in total.*

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| 2024-2032 | **Aerosol & Droplet Science CDT.** EPSRC, £1.10M. Local PI. |
| 2024-2029 | **Centre for Research in Biodetection Technologies (CRBT, E3).** Research England, £1.08M. Local PI. |
| 2020-2027 | **Digital Solutions Platforms and Toolkits.** NERC, £5.78M. Co-I. |
| 2024-2025 | **Talk Data to Me! LLMs for federated data discoverability.** ESRC, £133k. Co-I. |
| 2021-2023 | **Urban Observatories Open Data.** Department for Transport, £138k. PI. |
| 2021-2022 | **Towards Digital Twins of the Urban Environment in a net-zero world.** EPSRC, £216k. PI. |
| 2020-2024 | **Southern Ocean Clouds (SOC).** NERC, £493k. Local PI. |

## SELECTED PUBLICATIONS

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|---|---|
| 2025 | Zhang, H., Crawford, I. and **Topping**, D. et al. **Data-driven detection of nocturnal pollen fragmentation triggered by high humidity in an urban environment.** *Environmental Science & Technology*. doi:10.1021/acs.est.4c13905 |
| 2024 | Jay, C., Yu, Y., Crawford, I. ... **Topping**, D. et al. **Prioritize environmental sustainability in use of AI and data science methods.** *Nature Geoscience*. doi:10.1038/s41561-023-01369-y |

2023 Gledson, A., Lowe, D., Reani, M., **Topping**, D. et al. **A comparison of experience-sampled hay fever symptom severity across rural and urban areas of the UK.** *Scientific Reports 13, 3060.* doi:10.1038/s41598-023-30027-x

2021 **Topping**, D., Bannan, T. J., Coe, H., Evans, J. et al. **Digital twins of urban air quality: opportunities and challenges.** *Frontiers in Sustainable Cities 3, 786563.* doi:10.3389/frsc.2021.786563

2020 Huang, L. and **Topping**, D. **JlBox v1.0: a Julia-based mixed-phase atmospheric chemistry box model.** *Geoscientific Model Development.* doi:10.5194/gmd-2020-344

2019 McFiggans, G., Mentel, T. F., Wildt, J. ... **Topping**, D. and Kiendler-Scharr, A. **Secondary organic aerosol reduced by mixture of atmospheric vapours.** *Nature 565, 587-593.* doi:10.1038/s41586-018-0871-y

2022 **Topping**, D. and Bane, M. (Eds.). **Introduction to Aerosol Modelling: From Theory to Code.** *Wiley. ISBN 978-1-119-62565-0.* doi:10.1002/9781119625728

2009 **Topping**, D. **Thermodynamics of aqueous systems.** *In I. Colbeck (Ed.), Environmental Chemistry of Aerosols. Blackwell.* doi:10.1002/9781444305388.ch6

*Full list of more than 130 journal articles at https://loftytopping.github.io/publications.html and on ORCID.*

## OPEN-SOURCE SOFTWARE

**UManSysProp.** Original author and developer. Informatics suite for molecular property prediction and aerosol thermodynamics, used by more than 6,800 unique users on every major continent.

**PyBox.** Original author and developer. Automated atmospheric chemistry box-model generator.

**STRAPS.** Original author and developer. Prediction of aerosol mass spectra from molecular structure.

**JlBox.** Co-developer. High-performance Julia box model of atmospheric chemistry and physics.

## RESEARCH STUDENTS

**Main supervisor.** Pauline Nyakairu Karega, Hao Zhang, Samuel Hyman, Yuchen Zeng, Ximeng Kang, Congyu Zhao, Dinara Zhunissova, Maximillian Moss, Petroc Shelley, Simon Ruske, Elizabeth Forde.

**Co-supervisor.** Jessica Slater, Kathryn Fowler, Douglas Morrison, Emily Matthews, Caroline Dang, Simon O'Meara.

## TEACHING

**Natural Scientist's Toolkit.** Course lead for the programming component (sole lecturer) and the maths and physics component, around 140 first-year students.

**Short courses and summer schools.** Programming, machine learning, language models and aerosol modelling for doctoral and professional audiences.

**Data Science for Air Quality.** Course lead, MSc Pollution and Environmental Control.

## LEADERSHIP AND MANAGEMENT

2025-2028 **Director of Digital Worlds, Digital Futures.** University of Manchester

2022-2026 **Chair, Research IT Academic Advisory Group.** University of Manchester

2022-2025 **e-Research lead.** University of Manchester

## KNOWLEDGE EXCHANGE

2025-present **NVIDIA.** Collaborating with NVIDIA's Earth-2 team on generative AI for atmospheric science, following the Pollugen high-resolution air-quality work featured at the US-UK AI ecosystem event in London.

2025 **DSIT and Isambard-AI.** Demonstrated the value of the UK's national AI supercomputer for pollution modelling, released as a public case study.

## SELECTED INVITED TALKS

2026 **Beyond the Query: AI-enabled Environmental Science and Data Discovery.** University of Birmingham, Institute for Sustainability & Climate Action seminar series

2026 **Digital Solutions Hub.** UK Environment Agency Science Profession seminars

2024 **AI and the Environment (plenary).** UKEOF 2024 conference, UK Centre for Ecology and Hydrology

## CONFERENCE LEADERSHIP

2019, 2021 **Co-chair (2019) and Chair (2021).** International Aerosol Modelling Algorithms Conference (IAMA), UC Davis

2018 **Co-founder of the first Research Software Engineers working group in the European Geosciences Union**